

Practice Questions for 2024-10-01-DualGeneralLP

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1 Questions

1. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
 - a. The dual objective function $\xi(y, z)$ is unbounded below.
 - b. The dual objective function $\xi(y, z)$ is unbounded above.
 - c. The dual objective function $\xi(y, z)$ is bounded and has a finite minimum value.
 - d. The dual objective function $\xi(y, z)$ is bounded and has a finite maximum value.
 - e. We cannot conclude anything definitive about the dual objective function without further information.
2. Referring to Slide 1, we have the primal linear program (P): Maximize $\zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y) = c^T x - y^T (Ax - b)$, where y is the Lagrange multiplier. Explain the significance of the non-negativity constraint $y \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.
 - a. The constraint $y \geq 0$ ensures the Lagrangian is convex.
 - b. The constraint $y \geq 0$ is arbitrary and does not affect the dual problem.
 - c. The constraint $y \geq 0$ ensures the dual problem is a minimization problem.
 - d. The constraint $y \geq 0$ arises from the inequality constraints $Ax \leq b$ in the primal problem and is crucial for ensuring weak duality.
 - e. The constraint $y \geq 0$ is necessary for the Lagrangian to be differentiable.
3. Slide 3 shows the derivation of the dual problem from the Lagrangian. The inequality $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$ is derived. Explain the significance of the subsequent inequalities $\geq c^T$ and $\geq -c^T$ added under this inequality.
 - a. These inequalities are arbitrary and do not affect the final dual problem.
 - b. These inequalities ensure the dual problem is feasible.
 - c. These inequalities are necessary conditions for strong duality to hold.
 - d. These inequalities represent lower bounds on the expression, ensuring that the dual objective function is bounded below.

- e. These inequalities are derived from the objective function of the primal problem and are crucial for deriving the constraints of the dual problem.
4. In Slide 7, the final dual linear program (D) is presented as: Minimize $b^T y + u^T z$ subject to $A^T y + z = c$ and $y, z \geq 0$. Explain the economic interpretation of the dual variable z_i if the primal problem represents a production problem where x_i represents the quantity of product i produced, c_i is the profit per unit of product i , A represents resource consumption per unit of product, b represents the total available resources, and u represents the production capacity limits for each product.
- z_i represents the shadow price of resource i .
 - z_i represents the marginal profit increase from exceeding the production capacity limit for product i .
 - z_i represents the marginal cost of producing one more unit of product i .
 - z_i represents the shadow price of the production capacity limit for product i .
 - z_i represents the unit profit of product i .
5. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- The dual objective function $\xi(y, z)$ is unbounded.
 - The dual objective function $\xi(y, z)$ is bounded.
 - The dual objective function $\xi(y, z)$ is equal to zero.
 - The dual objective function $\xi(y, z)$ is infeasible.
 - We cannot conclude anything definitive about the dual objective function without further information.
6. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- The dual objective function $\xi(y, z)$ is unbounded below.
 - The dual objective function $\xi(y, z)$ is unbounded above.
 - The dual objective function $\xi(y, z)$ is equal to zero.
 - The dual objective function $\xi(y, z)$ is bounded and has a finite minimum value.
 - We cannot conclude anything definitive about the dual objective function $\xi(y, z)$ without further information.
7. Referring to Slide 1, we have the primal linear program (P): Maximize $\zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y) = c^T x - y^T (Ax - b)$, where y is the Lagrange multiplier. Explain the significance of the non-negativity constraint $y \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.

- a. The constraint $y \geq 0$ is arbitrary and does not affect the derivation of the dual problem.
 - b. The constraint $y \geq 0$ ensures that the Lagrangian remains a convex function.
 - c. The constraint $y \geq 0$ is necessary to guarantee the feasibility of the dual problem.
 - d. The constraint $y \geq 0$ arises from the requirement that the dual problem is a minimization problem.
 - e. The constraint $y \geq 0$ is a consequence of the non-negativity constraint $x \geq 0$ in the primal problem.
8. Slide 3 shows the derivation of the dual problem from the Lagrangian. The inequality $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$ is derived. Explain the significance of the subsequent inequalities $\geq c^T$ and $\geq -c^T$ added under this inequality.
- a. These inequalities are added to ensure the dual problem is feasible.
 - b. These inequalities are arbitrary and do not affect the final dual problem.
 - c. These inequalities are necessary to bound the objective function of the dual problem.
 - d. These inequalities are used to simplify the derivation and are not essential to the final result.
 - e. These inequalities arise from the non-negativity constraints on x^+ and x^- in the primal problem.
9. In Slide 7, the final dual linear program (D) is presented as: Minimize $b^T y + u^T z$ subject to $A^T y + z = c$ and $y, z \geq 0$. Explain the economic interpretation of the dual variable z_i if the primal problem represents a production problem where x_i represents the quantity of product i produced, c_i is the profit per unit of product i , A represents resource consumption per unit of product, b represents the total available resources, and u represents the production capacity limits for each product.
- a. z_i represents the shadow price of resource i .
 - b. z_i represents the marginal profit of product i .
 - c. z_i represents the shadow price of the production capacity limit for product i .
 - d. z_i represents the cost of producing one unit of product i .
 - e. z_i represents the total profit from selling all units of product i .
10. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- a. The dual objective function $\xi(y, z)$ is unbounded below.
 - b. The dual objective function $\xi(y, z)$ is unbounded above.
 - c. The dual objective function $\xi(y, z)$ is bounded and has a finite minimum value.
 - d. The dual objective function $\xi(y, z)$ is bounded and has a finite maximum value.
 - e. We cannot conclude anything definitive about the dual objective function without further information.

2 Answers

1. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
 - a. The condition $c^T - y^T A - z^T \neq 0$ implies that the dual objective function is unbounded below. In Slide 6, the Lagrangian is minimized with respect to x . If $c^T - y^T A - z^T \neq 0$, the term $(c^T - y^T A - z^T)x$ can be made arbitrarily small (approaching $-\infty$) by choosing an appropriate x , resulting in an unbounded dual objective function.
 - b. The dual problem is a minimization problem. An unbounded dual objective function would be unbounded below, not above.
 - c. If the condition $c^T - y^T A - z^T \neq 0$ holds, the dual objective function is unbounded, so it cannot have a finite minimum value.
 - d. The dual problem is a minimization problem, so it cannot have a finite maximum value.
 - e. The condition $c^T - y^T A - z^T \neq 0$ directly impacts the unboundedness of the dual objective function, as explained in Slide 6.
2. Referring to Slide 1, we have the primal linear program (P): Maximize $\zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y) = c^T x - y^T (Ax - b)$, where y is the Lagrange multiplier. Explain the significance of the non-negativity constraint $y \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.
 - a. The Lagrangian's convexity is not directly determined by the non-negativity of y .
 - b. The constraint $y \geq 0$ is not arbitrary; it's fundamental to the derivation and interpretation of the dual problem.
 - c. The dual problem being a minimization problem is a consequence of the primal problem being a maximization problem, not directly due to $y \geq 0$.
 - d. The non-negativity constraint $y \geq 0$ is directly linked to the inequality constraints $Ax \leq b$ in the primal problem. When deriving the dual, we consider the inequality $y^T (Ax - b) \leq 0$, which holds only if $y \geq 0$. This constraint is essential for establishing weak duality, ensuring that the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem.
 - e. The Lagrangian's differentiability is not directly related to the non-negativity of y .
3. Slide 3 shows the derivation of the dual problem from the Lagrangian. The inequality $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$ is derived. Explain the significance of the subsequent inequalities $\geq c^T$ and $\geq -c^T$ added under this inequality.
 - a. These inequalities are not arbitrary; they are essential for the derivation.
 - b. Feasibility of the dual problem is determined by the constraints, not these inequalities directly.
 - c. Strong duality is a consequence of the relationship between the primal and dual optimal values, not a direct result of these inequalities.
 - d. These inequalities do not directly ensure the dual objective function is bounded below; that is ensured by the constraints of the dual problem.

- e. The inequalities $\geq c^T$ and $\geq -c^T$ are not arbitrary additions. They are derived from the primal objective function, $c^T x = c^T(x^+ - x^-)$. To ensure that the inequality derived from the Lagrangian remains valid for all feasible x^+ , x^- , we must have the left-hand side bounded below by c^T (when $x^+ > 0$ and $x^- = 0$) and above by $-c^T$ (when $x^+ = 0$ and $x^- > 0$). This is crucial for deriving the constraints $A^T y + z \geq c$ in the dual problem.
4. In Slide 7, the final dual linear program (D) is presented as: Minimize $b^T y + u^T z$ subject to $A^T y + z = c$ and $y, z \geq 0$. Explain the economic interpretation of the dual variable z_i if the primal problem represents a production problem where x_i represents the quantity of product i produced, c_i is the profit per unit of product i , A represents resource consumption per unit of product, b represents the total available resources, and u represents the production capacity limits for each product.
- The shadow prices of resources are represented by the dual variables y_i .
 - While z_i relates to production capacity, it represents the shadow price, not the direct marginal profit increase from exceeding the limit.
 - The marginal cost of production is not directly represented by the dual variables in this formulation.
 - In this context, the dual variable z_i represents the shadow price of the production capacity limit for product i . It indicates the marginal increase in the objective function (total profit) if the production capacity limit for product i were increased by one unit. The dual constraint $A^T y + z = c$ reflects the trade-off between using resources (y) and exceeding capacity limits (z) to maximize profit.
 - The unit profit of product i is given by c_i in the primal problem.
5. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- The correct answer is A. If $c^T - y^T A - z^T \neq 0$, it means that the gradient of the Lagrangian with respect to x is not zero. In the process of deriving the dual, this condition leads to an unbounded dual objective function because the Lagrangian can be made arbitrarily small by choosing appropriate values of x .
 - Incorrect. The condition $c^T - y^T A - z^T \neq 0$ implies that the dual problem is unbounded, not bounded.
 - Incorrect. The value of the dual objective function is not necessarily zero; it's unbounded.
 - Incorrect. Infeasibility of the dual problem is not directly implied by this condition. The dual problem is unbounded.
 - Incorrect. The condition $c^T - y^T A - z^T \neq 0$ provides sufficient information to conclude that the dual objective function is unbounded.
6. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- The condition $c^T - y^T A - z^T \neq 0$ implies that the dual objective function is unbounded below. In the derivation of the dual, this term multiplies the primal variable x in the Lagrangian. If this

term is non-zero, the Lagrangian can be made arbitrarily small by choosing an appropriate value of x , leading to an unbounded dual objective function.

- b. The dual objective function is a minimization problem. If it were unbounded, it would be unbounded below, not above.
 - c. There's no reason to believe the dual objective function would be zero. The condition only tells us about boundedness.
 - d. The condition implies the opposite: the dual is unbounded.
 - e. The condition $c^T - y^T A - z^T \neq 0$ directly impacts the unboundedness of the dual objective function.
7. Referring to Slide 1, we have the primal linear program (P): Maximize $\zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The Lagrangian is given by $L(x, y) = c^T x - y^T (Ax - b)$, where y is the Lagrange multiplier. Explain the significance of the non-negativity constraint $y \geq 0$ in the context of the Lagrangian and the derivation of the dual problem.
- a. The constraint is not arbitrary; it's essential for the derivation and feasibility of the dual.
 - b. While the Lagrangian's properties are important, the non-negativity constraint on y is primarily about feasibility, not convexity.
 - c. The non-negativity constraint $y \geq 0$ is crucial for the feasibility of the dual problem. The dual problem is derived by minimizing the Lagrangian with respect to x and then maximizing the resulting expression with respect to y . The constraint $y \geq 0$ ensures that the dual problem remains a well-defined minimization problem with a feasible region. Without this constraint, the dual problem might be unbounded or infeasible.
 - d. The constraint $y \geq 0$ is not a direct consequence of the dual being a minimization problem; it's a consequence of the derivation process and the nature of the constraints in the primal problem.
 - e. While there's a relationship between the constraints in the primal and dual problems, the non-negativity of y is not a direct consequence of the non-negativity of x . It's a consequence of the Lagrangian's structure and the desire for a feasible dual problem.
8. Slide 3 shows the derivation of the dual problem from the Lagrangian. The inequality $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$ is derived. Explain the significance of the subsequent inequalities $\geq c^T$ and $\geq -c^T$ added under this inequality.
- a. Feasibility is ensured by the constraints $y \geq 0$ and $z \geq 0$, not these additional inequalities.
 - b. These inequalities are not arbitrary; they are essential for deriving a well-defined dual problem.
 - c. The inequalities $\geq c^T$ and $\geq -c^T$ are crucial for bounding the objective function of the dual problem. The original inequality involves both x^+ and x^- . To obtain a meaningful dual problem, we need to ensure that the expression is bounded from below. The inequalities $\geq c^T$ and $\geq -c^T$ provide these bounds, leading to the final form of the dual objective function and constraints.
 - d. These inequalities are not merely for simplification; they are fundamental to the derivation of the dual objective function.
 - e. While the non-negativity of x^+ and x^- is important, the inequalities $\geq c^T$ and $\geq -c^T$ are not a direct consequence of these constraints. They are a consequence of the need to bound the expression to derive a meaningful dual problem.

9. In Slide 7, the final dual linear program (D) is presented as: Minimize $b^T y + u^T z$ subject to $A^T y + z = c$ and $y, z \geq 0$. Explain the economic interpretation of the dual variable z_i if the primal problem represents a production problem where x_i represents the quantity of product i produced, c_i is the profit per unit of product i , A represents resource consumption per unit of product, b represents the total available resources, and u represents the production capacity limits for each product.
- Shadow prices are associated with resource constraints, not production capacity limits.
 - Marginal profit is already represented by c_i in the primal problem.
 - In this economic interpretation of the production problem, z_i represents the shadow price of the production capacity limit for product i . The dual variable z_i is associated with the constraint $x_i \leq u_i$, which limits the production of product i . The value of z_i indicates how much the optimal objective function value (total profit) would increase if the production capacity limit u_i were increased by one unit.
 - The cost of production is not directly represented by the dual variables.
 - The total profit from selling all units of product i is given by $c_i x_i$, not z_i .
10. Consider the general linear program (GLP) presented in Slide 2: Maximize $c^T x$ subject to $Ax \leq b$ and $x \leq u$. The Lagrangian is formed, incorporating Lagrange multipliers y and z . If, during the process of finding the dual, we determine that $c^T - y^T A - z^T \neq 0$, what can we definitively conclude about the dual objective function $\xi(y, z)$, and why?
- The condition $c^T - y^T A - z^T \neq 0$ implies that the dual objective function is unbounded below. In the derivation of the dual, this term multiplies the primal variable x in the Lagrangian. If this term is non-zero, the objective function can be made arbitrarily small by choosing appropriately large values of x , leading to an unbounded dual objective function.
 - The dual problem is a minimization problem. An unbounded dual would be unbounded below, not above.
 - If the term is non-zero, the dual is unbounded below.
 - The dual problem is a minimization problem. An unbounded dual would be unbounded below, not above.
 - The condition provides sufficient information to determine that the dual is unbounded below.